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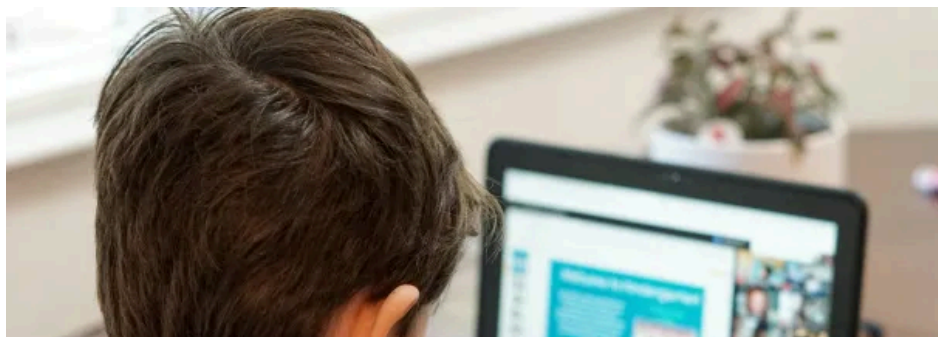
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Baumol's cost disease: long-term economic implications where machines cannot replace humans

Education faces a cost disease: despite AI advances, productivity stagnates. This piece explores the roots of this paradox and its implications for the future.

7 May 2025 - Last update: 27 October 2025



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Biographies

By Guido Appoldt, Shanghai Jiao Tong University, Shanghai, China

By Zhen Gong, Shanghai Creative Center of Arts and Culture, Shanghai, China

Education has long suffered from what economists term “Baumol’s cost disease,” a tendency for costs to rise faster than general inflation due to stagnant productivity in labor-intensive sectors (Kimball & Luke, 2018). In higher education this phenomenon is sometimes dubbed “Bowen’s curse” (Baumol, 2012). Essentially, teaching is time and labor intensive: a professor in the 2020s still needs to spend roughly the same amount of time to effectively teach or mentor a student as decades ago, just as *“it still takes four musicians nine minutes to perform Beethoven’s String Quartet in C minor, as it did in the nineteenth century”* (Kimball & Luke, 2018). In contrast, technology has dramatically boosted productivity in manufacturing and other sectors, enabling fewer workers to produce more output.

The crux of the issue is that education’s core service cannot

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alleviate or exacerbate cost pressures, why its impact is limited, and its long-term influence on the future of education.

Persistence of Cost Disease in Education

Baumol's cost disease theory holds that sectors requiring direct human input (like teaching) experience relatively low productivity growth because they cannot significantly speed up or substitute the human labor involved (Costa Nunes & Nunes, 2024). In education, *"the labor is itself the end product"* (Kimball & Luke, 2018), a teacher's presence and interaction is fundamental to the service. No matter how advanced technology becomes, most teaching tasks remain resistant to automation. As a recent analysis notes, jobs emphasizing interpersonal interactions (e.g. *face-to-face communication with learners*) are *"likely to be unaffected or not enabled by [AI]"*, and *"most teaching tasks ... feature prominently in this category"* (Elhussein et al., 2024). In other words, AI cannot conduct a parent-teacher conference, inspire a struggling student, or manage a lively classroom discussion with the same effectiveness as a human teacher.

"In other words, AI cannot conduct a parent-teacher conference, inspire a struggling student, or manage a lively classroom discussion with the same effectiveness as a human teacher. AI is able to automate certain education-related tasks, but these

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AI is able to automate certain education-related tasks, but these are generally auxiliary duties rather than the core act of teaching. For instance, AI systems can grade multiple-choice exams, compile learning materials, or analyze student performance data much faster than humans (Elhussein et al., 2024). Offloading such routine and administrative tasks does offer productivity gains as studies show teachers spend a large portion, often over 50%, of their work hours on non-teaching tasks (Elhussein et al., 2024). Automating these burdens could alleviate stress and slightly improve overall efficiency (Elhussein et al., 2024). AI is already being applied in this vein: for example, machine learning models have been developed to optimize resource allocation and recommend personalized learning paths in higher education, showing promising improvements in decision-making efficiency (Qian et al., 2025).

However, these efficiency gains do not translate into a proportional reduction in labor needs for the central teaching role. Whether a class uses AI tools or not, each additional group of students generally still requires a teacher's oversight, guidance, and feedback. Advanced AI like large language models can support lesson planning or offer tutoring, but human oversight remains necessary to ensure accuracy, appropriateness, and emotional support (Anderson, 2023). The World Economic Forum observes that while AI can assist with lesson preparation or grading, teachers must still handle *"the essential educational aspects of interpersonal interaction"* (Elhussein et al., 2024). In practice, AI tutors or

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"In short, Baumol's cost disease persists: despite new technologies, the cost of education tends to keep climbing relative to other sectors because teaching still requires intensive human involvement."

Sometimes, AI even *adds* new responsibilities: a U.S. Department of Education report emphasizes that teachers need to "*exercise judgment and control over the use of AI in education*" at all times (National Education Association, 2024). Similarly, UNESCO warns that aiming to eliminate human teachers betrays a "*fundamental misunderstanding of their essential social role in the learning process*" (UNESCO, 2021). Therefore, AI has not overturned the fundamentally labor-intensive nature of education, and productivity growth remains modest. As one commentary analogizing investigative journalism notes, these sectors "rely heavily on human labor and struggle to achieve significant productivity gains from technological advances" (Wellbrock, 2024). In short, Baumol's cost disease persists: despite new technologies, the cost of education tends to keep climbing relative to other sectors because teaching still requires intensive human involvement.

Long-Term Implications

If the cost disease in education continues unabated and AI provides only incremental relief, there are serious **long-run implications** for the educational system and society.

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increasing class sizes, reducing staff-student ratios, or asking teachers to take on heavier workloads (Baumol & Bowen, 1965; Throsby, 1996). This can dilute the attention each student receives and strain teachers. Many educators already feel overstretched: surveys indicate that excessive administrative duties and large class sizes are top stressors, detracting from time spent on “quality interactions with students” (Elhussein et al., 2024). If nothing is done, the trend could worsen, teachers might be supervising more students or juggling additional responsibilities such as IT management of AI tools without relief. Research in the economics of education suggests that talented individuals may also shun or leave the teaching profession if working conditions and pay remain unattractive (Elhussein et al., 2024). Losing experienced teachers or being unable to recruit high-caliber new teachers is directly harmful to educational quality. Thus, Baumol's disease can manifest as a quality disease: rising costs not met with rising budgets result in fewer resources per student and potentially a narrower curriculum, undermining the richness of education.

"Ultimately, society faces a choice: either accept a decline in educational talent and quality, or subsidize education more heavily. "

Baumol's cost disease also implies a growing wage gap between education and other sectors which could fuel

~~teacher shortages and require financial reallocation from the~~

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Over time, teacher pay needs to keep pace with wages in other graduate-level professions. Indeed, data show that in many countries teaching salaries lag significantly behind those of similarly educated workers (often by 10–30%) (Elhussein et al., 2024), making teaching a less attractive career financially. This gap is a direct manifestation of cost disease: without productivity gains, any wage increases in education translate straight into higher per-student costs (Kimball & Luke, 2018). If budgets don't rise accordingly, salaries stagnate and talented individuals may choose higher-paying industries, aggravating the teacher gap.

Ultimately, society faces a choice: either accept a decline in educational talent and quality, or subsidize education more heavily. In practical terms, that means channeling resources from productivity-rich parts of the economy through taxation or public spending into education (Baumol, 1996). Other industries' growth indirectly funds teacher pay raises and smaller class sizes. Baumol himself argued that as productivity (and wealth) grows in the overall economy, the society *can* afford these rising education costs, but only if we consciously decide to allocate the funds (Baumol, 2012). Without such reallocation, the funding shortfall will persist, and education could enter a vicious cycle of underinvestment. Without deliberate interventions, the public school and institutions will be affected most severely (Costa Nunes & Nunes, 2024), resulting in the exacerbation of educational inequalities, we could be “sleepwalking into a world that widens existing inequalities” as a result of AI as UNESCO

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"Baumol's cost disease serves as a sobering reminder that not all problems yield to technological quick fixes. Education, at its heart, is a person-to-person endeavor. Thus, even the most sophisticated AI will not radically increase throughput without risking the loss of human connection and quality. "

An unequal future is yet avoidable

Baumol's cost disease serves as a sobering reminder that not all problems yield to technological quick fixes. Education, at its heart, is a person-to-person endeavor. Thus, even the most sophisticated AI will not radically increase throughput without risking the loss of human connection and quality. The analysis above has illustrated why AI has so far failed to resolve the cost disease in education. Consequently, costs continue to rise, straining schools and governments, while the specter of declining quality and growing inequality looms if we respond poorly.

"The situation calls for bold recognition that education is a "stagnant" sector that nonetheless underpins all other progress. Rather than viewing the rising cost of education as a crisis to avoid, we should view it as a natural investment in our future, one that

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rich and mediocrity for the rest. Yet this outcome is not inevitable as long as the necessary policy interventions are taken. The situation calls for bold recognition that education is a “stagnant” sector that nonetheless underpins all other progress. Rather than viewing the rising cost of education as a crisis to avoid, we should view it as a natural investment in our future, one that can be managed and is affordable.

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Guido Appoldt is a postgraduate student at the School of Media and Communication, Shanghai Jiao Tong University. His works explore the intersections of emerging technologies and sustainable development.

Zhen Gong is a playwright at the Shanghai Creative Center of Arts and Culture, creating both artistic and analytical works that address contemporary social issues.

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